

Linear Regression with Linear Algebra

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Utilising linear algebra allows for parallel computation & therefore makes our model much easier to compute mathematically.

Our true, unknown model utilising linear algebra will therefore be:

$$Y = X\beta + \epsilon \quad \text{where:}$$

- Y – is a $n \times 1$ matrix (response matrix)
- X – is a $n \times p$ matrix (design matrix)
- β – is a $p \times 1$ matrix (coefficient matrix)
- ϵ – is a $n \times 1$ matrix (error matrix)

$$\begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & \dots & X_{p-1,1} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & & \vdots \\ 1 & X_{1n} & \dots & X_{p-1,n} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

Again, since we do not know the true coefficient matrix values, we use estimated values given data:

$$\hat{Y} = X\hat{\beta} + e$$

where:

- $\hat{\beta}$ – $p \times 1$ estimated coefficient matrix
- e – $n \times 1$ residual matrix

Solution to Equation

Each feature acts as a new dimension of data with each column of X acting as n -th dimensional vectors in a p -th dimensional space. Our target $\hat{Y}$ is a n -th dimensional vector that casts a “shadow” on the hyperplane made by X . We do this by minimising the perpendicular distance to this surface from Y .

Therefore we can find a closed form solution. From the previous proof we found that $E[\epsilon] = 0$, therefore in matrix form:

$$E[e] = \begin{bmatrix} 0 \\ \vdots \\ \vdots \\ 0_n \end{bmatrix}$$

Then we can find a formula for $E[\hat{Y}]$ and MSE:

$$E[\hat{Y}] = X\hat{\beta}$$

$$\begin{aligned} \text{MSE} &= f(\hat{\beta}) = (Y - \hat{Y})^2 \\ &= [Y - (X\hat{\beta})]^2 \\ &= [Y - (X\hat{\beta})]^T [Y - (X\hat{\beta})] \\ &= (Y^T - \hat{\beta}^T X^T)(Y - X\hat{\beta}) \\ &= Y^T Y - Y^T X\hat{\beta} - \hat{\beta}^T X^T Y + \hat{\beta}^T X^T X\hat{\beta} \\ &= Y^T Y - (X^T Y)^T \hat{\beta} - \hat{\beta}^T (X^T Y) + \hat{\beta}^T X^T X\hat{\beta} \end{aligned}$$

Derivation of Optimal Parameter

Therefore the optimal matrix of values $\hat{\beta}$ using partial derivatives is:

$$\begin{aligned}\frac{\partial f}{\partial \hat{\beta}} &= 0 - X^T Y - X^T Y + \left(\hat{\beta}^T X^T X\right)^T + \left(\hat{\beta}^T X^T X\right)^T \\ &= 0 - 2X^T Y + 2X^T X \hat{\beta}\end{aligned}$$

Setting the derivative to 0 to find the minimum:

$$\frac{2X^T Y}{2} = \frac{2\hat{\beta}^T X^T X}{2}$$

$$X^T Y = X^T X \hat{\beta}$$

$$(X^T X)^{-1} X^T Y = (X^T X)^{-1} (X^T X) \hat{\beta}$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

Where $X(X^T X)^{-1} X^T = H$ and H is our Hat matrix that finds our fitted value matrix $\hat{Y}$ from response vector Y .